

ARGUS

AI-Driven Energy Supply Chain Resilience for India

ET AI Hackathon 2.0 - Problem Statement 2

AI-Driven Energy Supply Chain Resilience for Import-Dependent Economies

GitHub: github.com/anshul23102/argus-energy-resilience

What this document is: a complete technical account of ARGUS - the problem it answers, the architecture, every engine and how it works, the technology stack, the data and assumptions behind every number, and how the system maps onto each item in the problem statement's evaluation focus and judging criteria.

1. Executive Summary

India sources roughly 88% of its crude oil from imports, with 40–45% of that volume transiting the Strait of Hormuz - a single chokepoint. India's Strategic Petroleum Reserves cover only about 9.5 days of national consumption. Traditional supply-chain planning tools cannot model geopolitical scenario impacts in real time, evaluate alternative procurement corridors dynamically, or orchestrate a coordinated response across refiners, logistics, and reserves.

ARGUS is that missing intelligence layer. It continuously monitors geopolitical and logistics risk from live news, models the cascading impact of specific disruption scenarios, and generates an executable procurement response - turning a reactive crisis into a managed, anticipated one.

Design philosophy: neuro-symbolic, not agent-does-everything

ARGUS deliberately keeps LLMs at the perception edges only - reading headlines, extracting structured signals, and writing the final briefing. Every risk score, every dollar, every barrel is computed by deterministic, auditable math: Bayesian updating, Monte Carlo simulation, and linear programming. The LLM never does the math. This is a conscious positioning against "wrap an LLM around everything" - the parts of a crisis response that must be trustworthy are computed, not generated.

2. System Architecture

ARGUS is a FastAPI (Python) backend and a Next.js 16 / React 19 / TypeScript frontend, with a shared, versioned assumptions file as the single source of truth for every parameter. The pipeline for a single response, end to end:

- **Watchtower** - the risk engine reads live evidence and produces a threat assessment (corridor and/or supplier).
- **Simulator** - the scenario engine runs a 1,000-path Monte Carlo simulation of the disruption's cascade.
- **Trader** - the procurement engine solves a linear program for the cheapest feasible replacement barrel mix.
- **Reservist** - the SPR engine schedules a strategic-reserve bridge and its post-crisis replenishment.
- **Briefer** - an LLM (with a deterministic template fallback) turns the structured output into a one-page briefing.

The full pipeline - from a scenario request to a complete, LLM-authored situation briefing - runs in under one second on a laptop, with every stage's wall-clock time reported in the response.

Data flow

Layer	What it does	Key files
Perception	GDELT + Google News RSS polling, LLM/rule-based headline extraction (corridor + supplier + severity)	news.py extractor.py

Layer	What it does	Key files
Risk	Bayesian corridor and supplier disruption scoring from decayed, corroborated evidence	<code>risk.py</code>
Scenario	Monte Carlo cascade simulation: chokepoint closure or OPEC+ production cut	<code>scenario.py</code>
Procurement	Linear program for cheapest feasible replacement barrels, tanker/port-constrained	<code>procurement.py</code>
Reserves	SPR drawdown scheduling and post-crisis replenishment window	<code>spr.py</code>
Orchestration	Ties every stage together, times each stage, drafts the briefing	<code>orchestrator.py</code>
Knowledge graph	Supplier-route-chokepoint-refinery relationships	<code>graph.py</code>
Frontend	War Room 3D globe, Corridor/Supplier Risk, Intelligence feed, Scenario Console, Network, Global, Assumptions, Sources	<code>app/(app)/*</code>

3. Core Modules

3.1 Geopolitical Risk Intelligence

A live-updating Bayesian engine scores disruption probability by **corridor** (7 chokepoints: Hormuz, Bab el-Mandeb, Suez, Malacca, Danish Straits, Turkish Straits, Cape of Good Hope) and independently by **supplier** (all 8 of India's crude sources). Every 15 minutes, ~500 headlines from GDELT and Google News are classified by an LLM (Gemini → Groq → deterministic rule-based fallback chain) into corridor, supplier, and severity ('rhetoric' through 'full closure'). Same-incident coverage is clustered so volume doesn't inflate the score. Each event decays the posterior probability with a 14-day evidence half-life, calibrated and validated against real historical episodes.

3.2 Disruption Scenario Modeller

Two shock types share one Monte Carlo engine (1,000 simulated paths, 90-day horizon):

- **Chokepoint closure** - a named strait closes at a chosen severity and duration; models bypass-pipeline recovery, refinery run-rate cuts, retail price pass-through, GDP and CAD impact.
- **OPEC+ emergency production cut** - a production-side shock across all 7 OPEC+-associated suppliers, no shipping route involved; the LP is barred from sourcing relief from any cutting supplier.
- **Power sector stress** (new) - models the indirect strain of a crude shock on India's grid: diesel backup-generator capacity squeezed by refinery run cuts, and gas-fired generation curtailed by cost stress. Explicitly not a direct oil-to-electricity effect, since India's grid is barely oil-fired (~0.2%).

Every scenario reports both a **managed** and an **unmanaged** trajectory side by side, so the value of ARGUS's own response is a measured number, not a claim.

3.3 Adaptive Procurement Orchestrator

A linear program (PuLP/CBC) finds the cheapest feasible replacement-barrel mix across every open (supplier, grade, route) combination, subject to real constraints:

- Supplier spare capacity (mb/d each supplier could realistically add within weeks)
- Refinery sour/sweet grade compatibility (India's slate is majority sour-configured)
- Closed chokepoints excluded entirely from candidate routes
- **Tanker availability** (new) - a per-route ceiling from voyage round-trip time and an available-tanker-fleet assumption, forcing route diversification instead of dumping all volume on one lane
- **Port congestion** (new) - an aggregate India-wide discharge-capacity ceiling using previously unused vlcc_capacity_mbbl / port_max_vlcc_per_day assumptions

Output is an executable order sheet: supplier, grade, route, volume, ETA, landed cost per barrel, and premium versus baseline.

3.4 Strategic Reserve Optimisation

Models optimal SPR drawdown against the supply gap (ISPRL Phase I, 5.33 MMT), and - new this round - the **replenishment window**: once relief cargoes arrive and a cooldown period passes, dedicated refill purchasing begins at a realistic injection rate until the reserve is back to full, with an exact day-count estimate.

3.5 Supply Chain Digital Twin

A live 3D globe (React Three Fiber) renders 22 refineries, ports, SPR sites, 8 supplier export terminals, and 10 shipping routes with risk-colored markers, plus a flat-map toggle. A NetworkX knowledge graph underlies supplier→route→chokepoint→refinery relationships, exportable to Neo4j.

4. Technology Stack

Layer	Technology
Backend	Python, FastAPI, NumPy, PuLP (CBC solver), NetworkX
LLM chain	Gemini → Groq → deterministic rule-based fallback (never a single point of failure)
Live data	GDELT + Google News RSS (news), yfinance (Brent/WTI/USDINR)
Frontend	Next.js 16 (App Router), React 19, TypeScript, Tailwind CSS
3D visualization	React Three Fiber / Three.js, custom globe↔flat-map morph shader
Testing	pytest - 37 backend regression tests across risk, scenario, procurement, SPR, extractor

5. Data, Assumptions & Transparency

Every numeric parameter ARGUS computes with - hazard priors, elasticities, capacity figures, response-speed assumptions - lives in a single versioned `assumptions.yaml`, each entry tagged with a source and a confidence level (high / medium / low / calibrated). Low-confidence values are not hidden - they are the explicit, testable, editable assumptions the evaluation focus asks for. The Assumptions page lets a judge change any parameter and re-run a scenario live to see the cascade move.

Confidence legend

- **high** - published hard data (e.g. VLCC capacity, retail pass-through arithmetic)
- **medium** - published estimate ranges (e.g. commercial stock days, refinery product slate)
- **low** - order-of-magnitude expert judgment, flagged in the UI (e.g. tanker fleet availability, power-sector transmission)
- **calibrated** - fitted and validated against real historical episodes (e.g. evidence half-life, severity likelihood ratios)

6. Backtesting & Validation

ARGUS replays 5 real historical disruption episodes - including the 2019 Abqaiq attack, the 2021 Ever Given Suez blockage, and 2023–24 Red Sea/Houthi escalation - through the exact same risk engine used live, with no lookahead: each event is only visible to the engine after its real-world timestamp. The engine's lead time and peak-risk accuracy are measured against what actually happened, and the core parameters (evidence half-life, severity likelihood ratios) were calibrated on one episode and validated out-of-sample on the others.

7. Mapping to the Problem Statement

7.1 Evaluation Focus

Evaluation Focus item	How ARGUS addresses it
Disruption signal detection lead time and accuracy	Backtested against 5 real historical episodes with no-lookahead replay; lead time reported per episode
Quality and executability of procurement alternatives	LP-generated order sheet: exact supplier, grade, route, volume, ETA, cost - not a suggestion
Scenario model fidelity, explicit and testable assumptions	Every parameter sourced, confidence-tagged, and live-editable on the Assumptions page
Geospatial evidence depth	22 refineries, ports, SPR sites, 8 suppliers, 10 routes on an interactive 3D globe with live risk coloring
End-to-end response time from signal to recommendation	Full pipeline (risk → scenario → procurement → SPR → briefing) instrumented and reported; consistently under 1 second

7.2 What You May Build - coverage

Suggested capability	Status
Geopolitical Risk Intelligence Agent	Built - live, corridor + supplier, continuous 15-min polling
Disruption Scenario Modeller	Built - chokepoint closure, OPEC+ cut, refinery/price/GDP/CAD/power-sector cascades
Adaptive Procurement Orchestrator	Built - LP with grade, spare-capacity, tanker, and port-congestion constraints
Strategic Reserve Optimisation Agent	Built - drawdown scheduling and replenishment window estimation
Supply Chain Digital Twin	Built - live 3D globe, knowledge graph, what-if scenario console

8. Repository

github.com/anshul23102/argus-energy-resilience

The repository includes the full backend and frontend source, the versioned assumptions file, 37 backend regression tests, and design/implementation documentation for every development phase under docs/superpowers/.